

Reducing bioavailable sex hormones through a comprehensive change in diet: the diet and androgens (DIANA) randomized trial.



High serum levels of testosterone and estradiol, the bioavailability of which may be increased by Western dietary habits, seem to be important risk factors for postmenopausal breast cancer. We hypothesized that an ad libitum diet low in animal fat and refined carbohydrates and rich in low-glycemic-index foods, monounsaturated and n-3 polyunsaturated fatty acids, and phytoestrogens, might favorably modify the hormonal profile of postmenopausal women. One hundred and four postmenopausal women selected from 312 healthy volunteers on the basis of high serum testosterone levels were randomized to dietary intervention or control. The intervention included intensive dietary counseling and specially prepared group meals twice a week over 4.5 months. Changes in serum levels of testosterone, estradiol, and sex hormone-binding globulin were the main outcome measures. In the intervention group, sex hormone-binding globulin increased significantly (from 36.0 to 45.1 nmol/liter) compared with the control group (25 versus 4%,; $P < 0.0001$) and serum testosterone decreased (from 0.41 to 0.33 ng/ml; -20 versus -7% in control group; $P = 0.0038$). Serum estradiol also decreased, but the change was not significant. The dietary intervention group also significantly decreased body weight (4.06 kg versus 0.54 kg in the control group), waist:hip ratio, total cholesterol, fasting glucose level, and area under insulin curve after oral glucose tolerance test. A radical modification in diet designed to reduce insulin resistance and also involving increased phytoestrogen intake decreases the bioavailability of serum sex hormones in hyperandrogenic postmenopausal women. Additional studies are needed to determine whether such effects can reduce the risk of developing breast cancer.